

**Virtual University of Pakistan**

**BT101**

**Ecology, Biodiversity & Evolution I**

**MCQs for Finals**

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# BT101 MCQ's FOR FINALS PREPRATION

- 1) The term ecology was invented by the Ernst Haeckel in
  - a) 1865
  - b) 1866**
  - c) 1867
- 2) A group of organisms of the same species that live in the same area at the same time called
  - a) Community
  - b) Ecosystem
  - c) Population**
- 3) Biomes are often defined by which factors
  - a) Biotic
  - b) Abiotic**
- 4) Prokaryotes can be divided into two domains
  - a) Viruses and bacteria
  - b) Archaea and bacteria**
  - c) Archaea and viruses
- 5) A catfish species in a pond usually shares the pond with other fish species, and with plants, insects, ducks, and many other species that make up the
  - a) Population
  - b) Specie
  - c) Community**
  - d) Ecosystem
- 6) The term "ecosystem" was first used by
  - a) Arthur Tansley**
  - b) Ernst Haeckel
- 7) Producers are
  - a) Heterotroph
  - b) Autotroph**
- 8) Which one is example of decomposers
  - a) Plants
  - b) Bacteria**
  - c) Animals
- 9) Which one is example of chemical components of environment
  - a) Air
  - b) Water
  - c) Oxygen**
  - d) Light
- 10) A pattern of living of species is called
  - a) Habitant
  - b) Niche**
- 11) horseshoe crab is the example of
  - a) Generalist specie**

- b) Native specie
  - c) Non-native specie
  - d) Specialist specie
- 12) One species of cockroach can survive being frozen for
- a) 24 hours
  - b) 30 hours
  - c) 32 hours
  - d) 48 hours**
- 13) In only a year, a single Asian cockroach and its offspring can add about how many new cockroaches to the world.
- a) 10 million**
  - b) 1 million
  - c) 7 million
  - d) 20 million
- 14) Those species that normally lives and thrives in a particular ecosystem called
- a) Generalist specie
  - b) Native specie**
  - c) Non-native specie
  - d) Specialist specie
- 15) Introduction of non native species is a major cause of-----
- a) Diversity
  - b) Evolution
  - c) Species loss**
  - d) Invasive species
- 16) Species that are deliberately or accidentally introduced into an ecosystem ?
- a) Native species
  - b) Non native species**
  - c) Indigenous species
  - d) Endemic species
- 17) Non native species also known as
- a) Exotic species
  - b) Invasive species
  - c) Alien species
  - d) All**
- 18) Species are introduced in the new environment by -- ways
- a) One
  - b) Two**
  - c) Three
  - d) Four
- 19) Intentional introduction is done for which cause?
- a) Economical gain
  - b) Neutralization societies

- c) **A and B**
  - d) None
- 20) When species are introduced by human vectors it is called
- a) **Unintentional introduction**
  - b) Intentional introduction
  - c) Both
  - d) None
- 21) What is the impact of non-native species?
- a) Diversity
  - b) Habitat loss
  - c) Ecosystem disturbance
  - d) **B and c**
- 22) Non-native species that have a potential to cause economic or environmental harm to humans are
- a) Endemic species
  - b) **Invasive species**
  - c) Endangered species
  - d) Extinct species
- 23) Grass carp is a great ----- source
- a) Carbohydrate
  - b) Lipids
  - c) **Protein**
  - d) Fats
- 24) In 1957, who imported wild African bees to help increase honey production?
- a) India
  - b) **Brazil**
  - c) Africa
  - d) China
- 25) Species that provide early warnings of damage to a community or an ecosystem ?
- a) Non native species
  - b) Endangered species
  - c) **Indicator species**
  - d) Keystone species
- 26) The presence or absence of trout species in water at temperatures within their range of tolerance is an indicator of
- a) Water level
  - b) Water nutrients
  - c) **Water quality**
  - d) Water abundance
- 27) Plants or lichens sensitive to heavy metals or acids in precipitation may be indicators of ?
- a) Water pollution
  - b) **Air pollution**
  - c) Land pollution

- d) All
- 28) A unique set of environmental qualities or characteristics found in a specific place is called
- a) Mixotrophs
  - b) Microclimate**
  - c) Microelements
  - d) Climate
- 29) Some species play a key and crucial role in maintaining the structure of a biological community are
- a) Indicator species
  - b) Keystone species**
  - c) Foundation species
  - d) Non-native species
- 30) Who studied the keystone role of the top-predator sea star?
- a) Robert hook
  - b) Robert Paine**
  - c) Robert grave
  - d) Robert Pike
- 31) Ecologist Robert Paine studied the keystone role of the top-predator sea star. Name of sea star was?
- a) Pisaster ochraceus**
  - b) Asteroidea
  - c) A and b
  - d) None
- 32) What is the keystone role of sharks in ecology?
- a) Remove plants from oceans
  - b) Remove live fishes from oceans
  - c) Remove injured and sick animals from oceans**
  - d) All of the above
- 33) The loss of keystone species can lead to
- a) Evolution
  - b) Population crashes
  - c) Extinctions of other species
  - d) B and c**
- 34) Specie that play an important role in structuring a community is called
- a) Keystone specie
  - b) Indicator specie
  - c) Foundation specie**
  - d) Non native specie
- 35) Which specie act as ecological engineers in streams and wetlands to create dams and ponds respectively used by other species
- a) Elephants
  - b) Fishes
  - c) Beavers**

- d) Bats
- 36) Which specie help to regenerate deforested areas?
- a) Elephants
  - b) Beavers
  - c) Bats and birds**
  - d) Lichens
- 37) Trophic means?
- a) Level
  - b) Food**
  - c) Web
  - d) Stage
- 38) Trophic structure means?
- a) Different feeding levels of organisms**
  - b) Different productivity levels of organisms
  - c) Different behaviors of organisms
  - d) All
- 39) The pyramid of number ignores the biomass of organisms and it also does not indicate the energy transferred or the use of energy by the groups involved.
- a) True**
  - b) False
  - c) Both
  - d) None
- 40) Ecological pyramid of number was advanced by
- a) Charles Elton**
  - b) Charles Baig
  - c) David
  - d) Boyles
- 41) The total biomass of the producers ingested by herbivores is----- than the total biomass of the herbivores in an ecosystem ?
- a) Less
  - b) More**
  - c) Larger
  - d) Smaller
- 42) How many kcal of energy is required by herbivores?
- a) 10,000 kcal**
  - b) 1000 kcal
  - c) 100 kcal
  - d) 10 kcal
- 43) How does the energy enters an ecosystem?
- a) Sunlight
  - b) Moon
  - c) Heat**
  - d) Nutrients

- 44) The highest level an organism occupies on a food chain is termed as
- a) Food level
  - b) Apex**
  - c) Trophic level
  - d) Food pyramid
- 45) Animals always live within a certain range of values, called the ----- for any environmental factor.
- a) Range of optimum
  - b) Tolerance range**
  - c) Range of temperature
  - d) Tolerance level
- 46) -----defines the conditions at which organisms are most successful in survival
- a) Range of optimum**
  - b) Tolerance range
  - c) Range of temperature
  - d) Tolerance level
- 47) When an abiotic factor becomes out of an organism tolerance range then it is said to be a limiting factor.
- a) Range of optimum
  - b) Tolerance range
  - c) Tolerance level
  - d) Limiting factor**
- 48) Response of an organism towards any abiotic factor is known as
- a) Phototaxis
  - b) Taxis**
  - c) Chemotaxis
  - d) None
- 49) a response of organism to light is called A response of organism to light is called
- a) Phototaxis**
  - b) Taxis
  - c) Chemotaxis
  - d) None
- 50) If the organism moves towards the stimulus, the taxis is ?
- a) Negative
  - b) Positive**
  - c) Neutral
  - d) Opposite
- 51) Each food chain ends with a
- a) Consumer
  - b) Carnivore
  - c) Top predator**
  - d) Quaternary consumer
- 52) The sequence of organisms through which energy moves in an ecosystem is a

- a) **Food chain**
  - b) Food web
  - c) Trophic level
  - d) Food pyramid
- 53) A ----- determines how nutrients and chemical energy moves through different trophic levels.
- a) **Food chain**
  - b) Food web
  - c) Trophic level
  - d) Food pyramid
- 54) Provide stability to ecosystem
- a) Food chain
  - b) **Food web**
  - c) Trophic level
  - d) Food pyramid
- 55) As energy flows from one organism to another energy is --
- a) Gained
  - b) **Lost**
  - c) Produced
  - d) Shared
- 56) Animals that have little or no natural enemies?
- a) Consumer
  - b) Carnivore
  - c) **Top predator**
  - d) Quaternary consumer
- 57) When an organism dies it is eventually eaten by -----
- a) Decomposers
  - b) **Detritivores**
  - c) Consumers
  - d) Plants
- 58) When a bear eats berries what is the trophic level of bear in food chain ?
- a) Primary producer
  - b) **Primary consumer**
  - c) Secondary consumer
  - d) Tertiary consumer
- 59) Food web is also termed as
- a) Resource system
  - b) Food system
  - c) **Consumer resource system**
  - d) Food chain
- 60) Autotrophic organisms that partially obtain organic matter from sources other than the atmosphere
- a) **Mixotrophs**
  - b) Microclimate

- c) Producers
  - d) Consumers
- 61) In deep sea hydrothermal ecosystem where there is no sunlight primary producers manufacture food through
- a) Photosynthesis
  - b) Chemosynthesis**
  - c) Heat
  - d) Boiling water
- 62) Mushrooms are?
- a) Producers
  - b) Decomposers**
  - c) Consumers
  - d) Plants
- 63) -----feed on large chunks of dead and decaying organic matter.
- a) Decomposer
  - b) Carnivores
  - c) Consumers
  - d) Scavenger**
- 64) Green plants use sunlight and water and make a type of sugar called
- a) Cellulose
  - b) Glucose**
  - c) Lactose
- 65) Bacteria use a process to make food called
- a) Photosynthesis
  - b) Chemosynthesis**
  - c) Both
- 66) organisms that derive their energy from the oxidation or breakdown of various inorganic or organic food substances in their environment
- a) Photoautotroph
  - b) Heterotroph
  - c) Chemoautotroph**
- 67) *Thiobacillus Ferrooxidans* is the scientific name of
- a) Onion
  - b) Cyanobacteria
  - c) Iron bacteria**
  - d) Raspberry
- 68) In terrestrial ecosystem plant capture --- solar energy falls on their leaves, to make food
- a) 1 %**
  - b) 10 %
  - c) 40 %
  - d) 27%
- 69) In open water, the dominant producers are
- a) Eutrophication

- b) Dinoflagellates
  - c) Phytoplankton**
  - d) Zooplankton
- 70) Cell wall of diatoms is made up of
- a) Lipids and proteins
  - b) Cellulose
  - c) Chitin
  - d) Silica**
- 71) Red tide is a common name for a phenomenon known as
- a) Microbes bloom
  - b) Algal bloom**
  - c) Bacterial bloom
- 72) What is energy currency of cell
- a) NADPH
  - b) ATP**
  - c) ADP
  - d) AMP
- 73) Which one is not example of heterotroph
- a) Animals
  - b) Algae
  - c) Plants**
  - d) Bacteria
- 74) Ominvores eat plants?
- a) Just fruits
  - b) Just vegetables
  - c) Eat both**
  - d) None of them
- 75) Butterflies are
- a) Primary consumers**
  - b) Secondary consumers
  - c) Tertiary consumers
  - d) None of them
- 76) A giraffe eat how many food each day
- a) 60 pounds
  - b) 75 pounds**
  - c) 65 pounds
  - d) 70 pounds
- 77) In deserts, secondary consumers that feed on smaller animals classified as
- a) Carnivoers**
  - b) Herbivores
  - c) Omnivores
- 78) Carnivorous fishes possess long stomach and short
- a) Bladder

- b) Liver
  - c) Gilles
  - d) Intestine**
- 79) Animals feed off the dead and break down the nutrients depositing to the soil.
- a) Producers
  - b) Decomposers**
  - c) Consumers
- 80) wood-decay fungi have evolved the enzymes necessary to decompose
- a) Chlorophyll
  - b) Lignin**
  - c) Cellulose
- 81) Detritivores include
- a) Bacteria
  - b) Fungi
  - c) Worms
  - d) All of above**
- 82) In a food chain, detritivores are
- a) Quaternary consumers
  - b) Tertiary consumers
  - c) Secondary consumers
  - d) Primary consumers**
- 83) Earthworm is a good example of
- a) Rocky detritivores
  - b) Terrestrial detritivores
  - c) Soil dwelling detritivores**
  - d) Both a and c
- 84) Detritivores and ----- are specific cases of consumer-resource systems.
- a) Fungi
  - b) Scavengers**
  - c) Viruses
  - d) Both a and b
- 85) The activity of animals feeding only on dead wood is called
- a) Sapro-xylophagous
  - b) Xylophagy
  - c) Sapro-xylophagy**
  - d) Detritivores
- 86) The eating of wood, whether live or dead, is known as
- a) Sapro-xylophagous
  - b) Xylophagy**
  - c) Sapro-xylophagy
  - d) Detritivores
- 87) Earth's life-support system consists of main spherical systems
- a) 2
  - b) 3
  - c) 4**
  - d) 5

- 88) The term atmosphere concern with
- a) Living organisms
  - b) Soil
  - c) Air**
  - d) Both a and c
- 89) Which of the following is greenhouse gas
- a) Oxygen
  - b) Nitrogen
  - c) Methane**
  - d) All of above
- 90) Hydrosphere is a .....at or near Earth's surface
- a) Continuous layer of water
  - b) Discontinuous layer of water**
  - c) Continuous Layer of hydrocarbons
  - d) Discontinuous layer of hydrocarbons
- 91) Most abundant substance on earth
- a) Rocks
  - b) Water**
  - c) Mountains
  - d) Soil
- 92) About ---- of water make up the oceans, lakes, streams, glaciers, and groundwater.
- a) 1.3 billion cubic kilometer
  - b) 1.4 billion cubic kilometer**
  - c) 1.5 billion cubic kilometer
  - d) billion cubic kilometer
- 93) Which part of earth system process shape of earth
- a) Atmosphere
  - b) Hydrosphere
  - c) Geosphere**
  - d) Biosphere
- 94) According to Christopher Columbus Earth is
- a) Complete sphere
  - b) Oblate spheroid**
  - c) Cubic
  - d) None of above
- 95) Radius of earth is ----from the Earth's center to the North Pole
- a) 6,237 kilometers
  - b) 6,347 kilometers
  - c) 6,357 kilometers**
  - d) 6,257 kilometers
- 96) Study of earth's shape is called
- a) Geosphere
  - b) Geodesy**
  - c) Satellites
  - d) Geography
- 97) The biosphere is the global sum of all ecosystems, is this statement
- a) True**

- b) False
- 98) Life created from living matter is termed as
- a) Biopoesis
  - b) Biogenesis**
  - c) Biology
  - d) Biosphere
- 99) Population ecology is a subfield of
- a) Ecology**
  - b) Biology
  - c) Both
  - d) None
- 100) Predation rate in summer is
- a) Low
  - b) High**
  - c) In between
- 101) Individuals are found in form of groups or clusters due to uneven distribution of resources
- a) Population growth
  - b) Clumping**
  - c) Uniform dispersion
  - d) Random dispersion
- 102) Ability of a population to grow under ideal conditions is called
- a) Biotic potential**
  - b) Low biotic potential
  - c) High biotic potential
- 103) The rate at which the population of a species would grow if it had unlimited resources.
- a) Intrinsic rate**
  - b) Growth rate
  - c) Population rate
  - d) Environmental resistance
- 104) The constraints that climate, food, space and other environmental factors place on a population are called
- a) Intrinsic rate
  - b) Growth rate
  - c) Population rate
  - d) Environmental resistance**
- 105) The maximum population that can survive indefinitely in a given environment
- a) Carrying capacity**
  - b) Growth rate
  - c) Population rate
  - d) Environmental resistance
- 106) What is the estimated carrying capacity of the earth ?
- a) 900 billion
  - b) 800 billion**
  - c) 700 billion
  - d) 500 billion

- 107) Exponential growth is also called  
a) Potential growth  
b) Exponential development  
c) Growth rate  
**d) Geometric growth**
- 108) Ability of individual to reproduce under ideal or optimum conditions.  
a) Potential growth  
**b) Reproduction potential**  
c) Growth rate  
d) Geometric growth
- 109) Which of the following factors influence the reproductive potential?  
a) Number of offspring  
b) Duration of the reproductive period  
c) Length of time it takes to maturity  
**d) All of above**
- 110) Together, biotic potential and environmental resistance determine the -----.  
a) Physical limit  
**b) carrying capacity (K )**  
c) Autophagy  
d) Intrinsic rate
- 111) The populations of individuals occurring within the center of the range.  
a) Peripheral population  
b) Edge population  
**c) Core population**  
d) Intrinsic population
- 112) -----are groups of individuals near the boundary of the range.  
a) Peripheral population  
**b) Edge population**  
c) Core population  
d) Both a and b
- 113) Ability of individuals to reproduce under ideal or optimum conditions.  
a) Exponential growth  
**b) Reproductive potential**  
c) Intrinsic rate  
d) Biotic potential
- 114) K and r change over time this statement is  
a) True  
**b) False**
- 115) Per capita growth rate involves  
a) The number deaths in a population  
b) The number births in a population  
**c) Both**  
d) None

- 116) Per capit means  
**a) Per individual**  
b) The number deaths in a population  
c) The number births in a population  
d) None
- 117) Meta population consists  
a) Different species  
b) Extinct species  
c) Both a and b  
**d) Same species**
- 118) Metapopulation is regulated by  
a) Extinction  
b) Colonization  
**c) Both**
- 119) Human population per unit area or volume is called  
a) Species  
b) Population  
**c) Population density**  
d) Community
- 120) Low densities may cause an extinction vortex and lead to further reduced fertility. This is called  
**a) Allee effect**  
b) Greenhouse effect  
c) Low population  
d) Population density
- 121) Density independent factors depend on the population density. This statement is  
a) True  
**b) False**
- 122) Density dependent factors depend on the population density. This statement is  
**a) True**  
b) False
- 123) What is the percentage of world's population living in less developed countries.  
a) 90 %  
**b) 80%**  
c) 70%  
d) 85%
- 124) The mixture of gases that surrounds the solid earth is called -----  
a) Surroundings  
b) Environment  
**c) Atmosphere**
- 125) Atmosphere is divided into ---- layers.  
a) 3  
b) 4

- c) 5  
d) 6
- 126) Nitrogen comes on which number in the abundance in the milky way.  
**a) 7th**  
b) 6th  
c) 5th  
d) 4th
- 127) Nitrogen was discovered in  
**a) 1772**  
b) 1779  
c) 1777  
d) 1778
- 128) Water vapor is ---- than air  
**a) Lighter**  
b) Heavier  
c) Same weights
- 129) ---- % of earth's surface is covered with water.  
a) 97  
**b) 71**  
c) 3  
d) 93
- 130) Lake ecosystem is example of  
a) Lotic ecosystem  
**b) Lentic ecosystem**
- 131) Lentus means -----  
a) Heavy  
b) Lentic  
**c) Sluggish**  
d) washed
- 132) Lotus means  
a) Heavy  
b) Lentic  
c) Sluggish  
**d) washed**
- 133) The study of fresh water is called  
**a) Limnology**  
b) Ecology  
c) Freshwater ecosystem  
d) Lotic water bodies
- 134) A freshwater body in which water flows out of the ground  
a) Stream  
b) River  
c) Pond

- d) Spring**
- 135) The biological habitat in the immediate vicinity of a stream is called-----
- a) Riparian zone**  
b) Limnetic zone  
c) Profundal zone  
d) Littoral zone
- 136) The study of streams and waterways in general is known as -----
- a) Limnology  
**b) Surface hydrology**  
c) Ecology  
d) Freshwater bodies
- 137) Nutrients and levels of a stream depends on stream's
- a) Water  
b) Springs  
c) Biomass  
**d) Watershed**
- 138) The scientific study of rivers is called
- a) Potamology**  
b) Limnology  
c) Hydrology
- 139) The study of inland water in general is called
- a) Potamology  
**b) Limnology**  
c) Hydrology
- 140) Which zone contains river which empty into the ocean
- a) Transition zone  
**b) Floodplain zone**  
c) Source zone
- 141) Lotic water bodies posses how many zones ?
- a) 1  
b) 2  
**c) 3**  
d) 4
- 142) Lentic water bodies posses how many zones ?
- a) 1  
b) 2  
c) 3  
**d) 4**
- 143) How many types of producers are present in littoral zone ?
- a) 3  
**b) 2**  
c) 1  
d) 4

- 144) Profundal zone has the-----water (4°C) in the lake in winter and ----- water in summer.  
a) Coldest, warmest  
**b) Warmest, coldest**  
c) Cold, coldest
- 145) Light can penetrate in profundal zone ?  
a) True  
b) False
- 146) Organisms living in the benthic zone are called benthos ?  
a) True  
**b) False**
- 147) What are the examples of benthic invertebrate community ?  
a) Crustaceans  
b) Polychaetes  
**c) Both**  
d) None
- 148) Overturn occurs in  
a) Winter and summer  
b) Summer and spring  
**c) Spring and fall**  
d) Fall and winter
- 149) Oligotrophic lakes are mostly found in ---- regions  
a) Warm  
**b) Cold**  
c) Both
- 150) Igneous rocks and granitic bedrock's underlain are present in  
**a) Oligotrophic lakes**  
b) Eutrophic lakes  
c) All lakes
- 151) What is the color of eutrophic lakes  
a) Murky brown  
b) Green  
c) Yellow  
**d) A and B**
- 152) State of hypoxia is present in  
a) Oligotrophic lakes  
b) Eutrophic lakes  
**c) Cultural eutrophication**  
d) Hypertrophic lakes
- 153) Wetlands play which of the following roles  
a) Water purification  
b) Flood control  
c) Carbon sink  
d) Shoreline stability

- e) All of above**
- 154) Prairie potholes is example of  
**(a) Wetland**  
b) Lake  
c) Eutrophic lake  
d) Oligotrophic lakes
- 155) Rice is the diet of how many people ?  
a) 4 billion people  
b) 5 billion  
c) 6 billion  
**d) 3 billion**
- 156) What is the salinity of brackish water?  
a) 0.4- 30 ppt  
b) 0.5-29 ppt  
**c) 0.5-30 ppt**
- 157) Which is the world's largest lake  
a) North sea  
b) Baltic sea  
**c) Caspian sea**
- 158) -----are formed where Pleistocene glaciers deepened and widened existing river valleys  
**a) Fjords**  
b) Estuary  
c) Bar built estuary  
d) Coastal plain estuary
- 159) ----- are formed by subsidence or land cut off from the ocean by land movement associated with faulting, volcanoes, and landslides.  
a) Fjords  
**b) Tectonic estuary**  
c) Bar built estuary  
d) Coastal plain estuary
- 160) -----are found in place where the deposition of sediment has kept pace with rising sea level  
a) Fjords  
b) Estuary  
**c) Bar built estuary**  
d) Coastal plain estuary
- 161) Drowned river valleys are also known as -----  
a) Fjords  
b) Estuary  
c) Bar built estuary  
**d) Coastal plain estuary**
- 162) A productive waterlogged area present between land and open sea is known as-----  
a) Fjords

- b) Estuary
  - c) Bar built estuary
  - d) Coastal wetlands**
- 163) ----- of all Great Lakes fish use coastal wetlands for spawning and reproduction.
- a) 2/3rd**
  - b) 3/4th
  - c) 1/2
  - d) 3/5
- 164) Around-----of tropical coastlines are mangrove-lined.
- a) 20%
  - b) 40%
  - c) 70%**
  - d) 60%
- 165) -----are waterlogged Peatlands in old lake basins or depressions in the landscape.
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs**
  - d) Marshes and swamp
- 166) Flooded depressions in grasslands or forests, and are often only wet in winter and early spring.
- a) Mangrove swamp
  - b) Shallow lakes and ponds**
  - c) Bogs
  - d) Marshes and swamp
- 167) The carnivorous pitcher plant is an example of ?
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs**
  - d) Marshes and swamp
- 168) ----- are unsuitable for agriculture, forestry or development.
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs**
  - d) Marshes and swamp
- 169) Provide habitat for species, including moose, black bear, lynx, snowshoe hare and mink.
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs**
  - d) Marshes and swamp
- 170) Also known as palustrine wetlands.
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs

- d) Marshes and swamp**
- 171) Dominated by floating-leafed plants like water lilies and duckweed.
- a) Mangrove swamp
  - b) Shallow lakes and ponds
  - c) Bogs
- d) Marshes and swamp**
- 172) Diversity which includes the variation in both terrestrial and aquatic ecosystems.
- a) Ecological diversity**
  - b) Functional diversity
  - c) Species diversity
  - d) Genetic diversity
- 173) Diversity which reflects the biological complexity of an ecosystem.
- a) Ecological diversity
  - b) Functional diversity**
  - c) Species diversity
  - d) Genetic diversity
- 174) Species diversity takes into account.
- a) Species richness
  - b) Species variation
  - c) Species evenness
  - d) Option a and c**
- 175) Serves as a way for populations to adapt to changing environments.
- a) Ecological diversity
  - b) Functional diversity
  - c) Species diversity
  - d) Genetic diversity**
- 176) The part of ocean that is deep and away from the coastal zone is termed as
- a) The coastal zone
  - b) Open sea**
  - c) Ocean bottom
- 177) -----also known as midnight zone
- a) Euphotic zone
  - b) Bathyal zone**
  - c) Abyssal zone
- 178) The Bathyal zone extends from a depth of.
- a) 1000 to 4000 meters**
  - b) 1000 to 5000 meters
  - c) 2000 to 4000 meters
  - d) 3000 to 6000 meters

- 179) Primary production does not exist in.  
a) Euphotic zone  
b) Bathyal zone  
c) Abyssal zone  
**d) Both b and c**
- 180) The abyssal zone depth is ?  
a) 1000 to 4000 meters  
b) 1000 to 5000 meters  
c) 2000 to 4000 meters  
**d) 4000 to 6000 meters**
- 181) The abyssal zone has temperatures around.  
a) 1 °C to 3 °C  
**b) 2 °C to 3 °C**  
c) 5 °C to 3 °C  
d) 2 °C to 6 °C
- 182) Showers of dead and decaying organisms drifting downward from upper lighted levels of the ocean.  
a) Abyssal zone  
**b) Ocean floor**  
c) Marine snow
- 183) Take in large amount of water.  
**a) Filter feeders**  
b) Deposit feeders  
c) Detritivore  
d) Scavengers
- 184) The overall economic benefit from marine coastal ecosystems is over \$---- trillion per year.  
a) 13  
**b) 12**  
c) 19  
d) 20
- 185) A microscopic swimming and free floating organisms in water are termed as  
**a) Plankton**  
b) Nekton  
c) Benthos  
d) Decomposers
- 186) Ultra plankton are responsible for ---- of the primary productivity near the ocean surface.  
a) 90%  
b) 50%

- c) **70%**  
d) 10%
- 187) This class of organisms includes active swimmers that act as a consumer.  
a) Plankton  
b) **Nekton**  
c) Benthos  
d) Decomposers
- 188) Oysters and sea stars are?  
a) Plankton  
b) Nekton  
c) **Benthos**  
d) Decomposers
- 189) The polyps belong to a group of animals known as Cnidaria, which also includes.  
a) Sea anemones  
b) Jellyfish  
c) **Both**  
d) Decomposers
- 190) Most modern coral reefs are less than-----years old.  
a) **10,000**  
b) 20,000  
c) 30,000  
d) 12,000
- 191) Coral reefs are important for  
a) Food  
b) Medicine  
c) Recreation  
d) Breeding grounds  
e) **All of above**
- 192) The area of shoreline between low and high tides is called -----.  
a) intertidal zone  
b) Internal zone  
c) Compensation zone  
d) Rocky shore
- 193) Which of the followings is called natural laboratory ?  
a) intertidal zone  
b) **Internal zone**  
c) Compensation zone  
d) Rocky shore
- 194) Loose deposits of sand, gravel or shells that cover the shoreline in many places.  
a) Rocky shore  
b) **Sandy shore**  
c) Beaches  
d) B and c

- 195) The movement of rocks and minerals by agents such as water, ice, snow, wind, waves and gravity and then being transported and deposited in other locations.
- a) **Weathering**
  - b) Erosion
  - c) Both
  - d) None
- 196) Weathering involves no movement this statement is
- a) **True**
  - b) False
- 197) Chemical reactions that cause changes in the minerals is the working of
- a) Chemical weathering
  - b) Mechanical weathering
  - c) Biological weathering
- 198) Temperature plays important in which of the weathering processes?
- a) **Chemical weathering**
  - b) Mechanical weathering
  - c) Biological weathering
- 199) Sub-microscopic clay particles play important role of
- a) Aeration
  - b) Drainage
  - c) Chemically active
  - d) Bind with plant and water nutrients
  - e) **C and D**
- 200) Largest particles that are sand particles play role in
- a) Aeration
  - b) Drainage
  - c) Chemically active
  - d) Bind with plant and water nutrients
  - e) **A and B**
- 201) Pedalfer soil has following characteristics.
- a) Acidic soil, found in wet areas, less leaching activity
  - b) Non-Acidic soil, found in wet areas, less leaching activity
  - c) **Acidic soil, found in wet climate, leaching activity**
  - d) Acidic soil, found in dry areas, less leaching activity
- 202) Which soil is rich in calcium carbonate and has low soil organic matter.
- a) Pedalfer soil
  - b) **Pedocal soil**
  - c) Laterite soil
  - d) Caliche
- 203) -----promotes growth of desert shrubs and small and medium grasses.
- a) Pedalfer soil
  - b) **Pedocal soil**
  - c) Laterite soil

- d) Caliche
- 204) A special type of Pedocal that is bright red and forms in tropical and subtropical climates with heavy rain and high temperature.
- a) Pedalfer soil
  - b) Pedocal soil
  - c) Laterite soil**
  - d) Caliche
- 205) Layer generally forms above the mineral soil or occurs in an organic soil profile.
- a) O-horizon**
  - b) A-horizon
  - c) B-horizon
  - d) C-horizon
- 206) Present in grassland and absent in forest for O-horizon this statement is
- a) True
  - b) False**
- 207) The topmost mineral horizon, often referred to as the 'topsoil'.
- a) O-horizon
  - b) A-horizon**
  - c) B-horizon
  - d) C-horizon
- 208) A-horizon has darker color due to
- a) Translocation
  - b) Erosion
  - c) Both**
- 209) Which horizon consists of large amount of minerals and organic matter?
- a) O-horizon
  - b) A-horizon**
  - c) B-horizon
  - d) C-horizon
- 210) Which horizon is predominantly the surface layer of many soils in grasslands and agricultural lands.
- a) O-horizon
  - b) A-horizon**
  - c) B-horizon
  - d) C-horizon
- 211) Silicate clays, iron (Fe) and aluminum (Al) oxides, and organic material are found in which horizon?
- a) O-horizon
  - b) A-horizon
  - c) B-horizon**
  - d) C-horizon
- 212) Which horizon is called as subsoil ?
- a) O-horizon

- b) A-horizon
  - c) B-horizon**
  - d) C-horizon
- 213) Which horizon lack the pedological development ?
- a) O-horizon
  - b) A-horizon
  - c) B-horizon
  - d) C-horizon**
- 214) Which horizon is also known as the saprolite, it is unconsolidated, loose parent material.
- a) O-horizon
  - b) A-horizon
  - c) B-horizon
  - d) C-horizon**
- 215) Humans normal body temperature is
- a) 98 F
  - b) 98.6 F**
  - c) 98.2 F
  - d) 98.5 F
- 216) What is the important function of temperature ?
- a) Metabolism
  - b) Seed germination
  - c) Opening and closing of flowers
  - d) Seasonal falling of tree leaves
  - e) All**
- 217) What is the range of healthy and normal body temperature ?
- a) 98°F (37°C) and 100°F (37°C).
  - b) 98°F (47°C) and 100°F (37.8°C).
  - c) 98°F (37°C) and 90°F (37°C).
  - d) 98°F (37°C) and 100°F (37.8°C).**
- 218) Hypothermia can cause
- a) Brain damage
  - b) Heart attack**
  - c) Both
- 219) Hyperthermia can cause
- a) Brain damage**
  - b) Heart attack
  - c) Both
- 220) Heat is released through skin from blood in which of processes
- a) Sweating
  - b) Vasodilation**
  - c) Vasoconstriction
  - d) Thermogenesis
- 221) Shivering is the function of ?

- a) Sweating
  - b) Vasodilation
  - c) Vasoconstriction
  - d) Thermogenesis**
- 222) Which of the following is the characteristic of ectotherm ?
- a) Low metabolic rate
  - b) Generate body heat from external environment
  - c) Body insulation is very poor
  - d) All of above**
- 223) Short period of hypothermia in endotherms is called
- a) Torpor**
  - b) Hibernation
  - c) Compensation period
  - d) None
- 224) Diurnal animals show inactivity in
- a) Daytime
  - b) Night time**
  - c) Both
- 225) Nocturnal animals show inactivity in
- a) Daytime**
  - b) Night time
  - c) Both
- 226) Which of the following is characteristic of endotherms ?
- a) Low rate of metabolism
  - b) Cannot generate heat from body cellular activities
  - c) Body insulation by fur or feather**
  - d) All
- 227) A state of decreased physiological activity in an animal, usually by a reduced body temperature and metabolic rate.
- a) Torpor**
  - b) Hibernation
  - c) Hypothermia
  - d) Hyperthermia
- 228) Whose arousal relies on a heat-producing brown adipose tissue as a mechanism to accelerate rewarming.
- a) Marsupial mammals
  - b) Eutherial mammals**
  - c) None
  - d) Both
- 229) Marsupial arousal does not depends on brown adipose tissues this statement is
- a) False
  - b) True**
- 230) A state of inactivity and metabolic depression in endotherms.

- a) Torpor
  - b) Hibernation**
  - c) Hyperthermia
  - d) All
- 231) Which fat gives energy to animal's brain and heart ?
- a) Whit fat
  - b) Brown fat**
  - c) Food
  - d) None
- 232) Which of the following is true hibernator
- a) Rodent
  - b) Shrews
  - c) Bats
  - d) All of them**
- 233) ----- is characterized by change in brain activity.
- a) Hibernation
  - b) Torpor
  - c) Sleep**
- 234) Inactivity and a lowered metabolic rate that is entered in response to high temperatures and arid conditions is known as
- a) Hibernation torpor
  - b) Aestivation**
  - c) Hyperthermia
- 235) Which of the following show the process of aestivation
- a) Gastropods
  - b) Insects
  - c) Crustaceans
  - d) All of them**
- 236) Hedgehogs show the process of ----- in dry season.
- a) Torpor
  - b) Hibernation
  - c) Aestivation**
  - d) Inactivity
- 237) Chuckwalla can tolerate the temperature up to
- a) 90° C
  - b) 50° C
  - c) 30° C
  - d) 40° C**
- 238) Birds orient themselves in a specific direction depending on the slight difference of----- reflected from object such as trees.
- a) Visible light
  - b) UV light**
  - c) IR light

- d) No light
- 239) Light which is absorbed mostly in aquatic ecosystem is
- a) Red and yellow light
  - b) Blue and red light**
  - c) Red and green light
  - d) Yellow and blue light
- 240) During the period from-----the Southern Hemisphere receives--- than 12 hours of sunlight.
- a) March to September, less**
  - b) April to October, less
  - c) March to April, less
  - d) September to March, more
- 241) Spinach, barley, wheat, radish, clover they are examples of
- a) Long day plants**
  - b) Short day plants
  - c) Day neutral plants
- 242) Chrysanthemum sp., Datura stramonium etc they are examples of
- a) Long day plants
  - b) Short day plants**
  - c) Day neutral plants
- 243) Tomato, maize they are examples of
- a) Long day plants
  - b) Short day plants
  - c) Day neutral plants**
- 244) Electromagnetic radiation is a
- a) Parallel wave
  - b) Transverse wave**
  - c) Longitudinal wave
  - d) Sound wave
- 245) IR radiation frequency is ---- than visible light.
- a) Less
  - b) More**
  - c) Equal to
- 246) Wavelength of gamma rays is ----- than X - rays.
- a) Greater
  - b) Lesser**
  - c) Equal to
- 247) Which of the following is the right sequence of increasing frequencies and decreasing wavelengths?
- a) radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays.**
  - b) radio waves, , infrared radiation, visible light, microwaves, X-rays and gamma rays, ultraviolet

- c) microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays, radio waves
- 248) The physics of electromagnetic radiation
  - a) Physics
  - b) Electrodynamics**
  - c) Electromagnetism
  - d) Electricity
- 249) The physical phenomenon associated with the theory of electrodynamics.
  - a) Physics
  - b) Electrodynamics
  - c) Electromagnetism**
  - d) Electricity
- 250) -----makes up most of the spectrum of electromagnetic radiation emitted by the sun.
  - a) IR light
  - b) Visible light**
  - c) UV light
  - d) X-ray light
- 251) EMR of small wavelength have
  - a) More energy**
  - b) Less energy
  - c) Equal energy
- 252) The relative amount of mechanical work that can be generated in a theoretical conversion process.
  - a) Conservation of energy
  - b) Energy
  - c) Quality of energy**
  - d) Quantity of energy
- 253) Nuclear fission is an example of
  - a) Low quality energy
  - b) High quality energy**
  - c) Neutral energy
- 254) Less quality energy has
  - a) Less entropy**
  - b) More entropy
  - c) No entropy
- 255) Quaternary consumer has no predator. This is
  - a) False
  - b) True**
- 256) -----is an important tool for the determination of community structure of a specific place.
  - a) Species richness
  - b) Species composition**
  - c) Species variation
  - d) Species competition

- 257) Which gives the character of vegetation of a particular site.  
**a) Ecological indicators**  
b) Management indicators  
c) Both  
d) none
- 258) Relative abundance of desirable forage species at a site influences the  
**a) Carrying capacity**  
b) Vegetation  
c) Both  
d) None
- 259) Wildlife is influenced by  
a) Abundance of forage species  
b) Species that provide source of shelter food  
**c) Both**
- 260) The number of species in an area is called  
a) Species abundance  
**b) Species richness**  
c) Species variation  
d) Species evenness
- 261) The percent composition of an organism of a particular kind relative to the total number of organisms in the area.  
**a) Species abundance**  
b) Species richness  
c) Species variation  
d) Species evenness
- 262) Relative species abundance and species richness describe key elements of -----  
a) Ecosystem  
**b) Biodiversity**  
c) Variations  
d) Community
- 263) ----- is a measure of the relative abundance of the different species that contribute to the richness of an area.  
a) Species abundance  
b) Species richness  
c) Species variation  
**d) Species evenness**
- 264) Species dominance has ----- relation with species diversity.  
a) Direct  
**b) Inverse**  
c) No relation
- 265) Tropical rain forests are home to ---- species  
a) 60%  
**b) 50%**

- c) 40%
- d) 20%
- 266) Succession that is driven by living components of environment is
  - a) **Autogenic succession**
  - b) Allogenic succession
  - c) Primary succession
  - d) Secondary succession
- 267) Which is not the benefit of succession ?
  - a) Development of an ecosystem.
  - b) Sustainability of ecosystem.
  - c) Recovery of ecosystem from damage caused by humans or any other natural disaster.
  - d) **Imbalance of nature**
- 268) ----- is a process of gradual changes that causes the plants, animals, fungi and microorganism to become established in an area.
  - a) Biological succession
  - b) **Ecological succession**
  - c) Primary succession
  - d) Allogenic succession
- 269) On sand dunes ----- is the first species to grow.
  - a) Grasses
  - b) **Beach grass**
  - c) Trees
  - d) Tiny shrubs
- 270) Candidates for secondary succession include
  - a) Abandoned farmland
  - b) Burned or cut forests
  - c) Heavily polluted streams
  - d) Flooded lands
  - e) **All of above**

# Finals ( topic 85-150)

- 271) Open self-organizing living things that interact with their environment.
  - a) Open systems
  - b) **Living systems**
  - c) Closed systems
  - d) Positive feedback
- 272) All living systems from a cell to the biosphere are ----- systems.
  - a) **Dynamic**
  - b) Static
  - c) Both

- 273) Thermostat is an example of  
a) Positive feedback  
b) Closed system  
c) **Negative feedback**  
d) None
- 274) The ability of a living system to survive in conditions of moderate disturbances.  
a) Persistence  
b) Inertia  
c) **Both**  
d) None
- 275) Grassland and forest show stability by  
a) Resilience  
b) Inertia  
c) Persistence  
d) **Both b and c**
- 276) The ability of a living system to be restored through secondary succession after a moderate disturbance.  
a) **Resilience**  
b) Inertia  
c) Persistence  
d) Both b and c
- 277) Leopards and lions show ----- competition.  
a) Interspecific  
b) **Interspecific**  
c) Both
- 278) The resource must be limited for competition to occur.  
a) **True**  
b) False
- 279) Interspecific competition is a stronger force than intraspecific competition?  
a) True  
b) **False**
- 280) ----- studied five species of warblers that all used the same caterpillar prey.  
a) **Robert MacArthur**  
b) Robinson  
c) Aristotle  
d) Albert Bernhard
- 281) ----- is used to describe cases where two (or more) species reciprocally affect each other's evolution.  
a) **Coevolution**  
b) Coexistence  
c) Competition  
d) All of above
- 282) Flowers pollinated by hummingbirds, are often -----

- a) Circular and blue
  - b) Irregular and white
  - c) Tubular and red.**
  - d) Of any shape
- 283) The rue produces ----- which repel plant-eating insects.
- a) Olive oils
  - b) Resistance oils
  - c) Etheric oils**
- 284) Who used the word symbiosis to describe the mutualistic relationship in lichens?
- a) 1877 Albert Bernhard Frank.
  - b) 1866 Albert Bernhard.
  - c) 1899 Bernhard Frank.
- 285) Which is a close and often long-term interaction between two different biological species?
- a) Mutualism
  - b) Commensalism
  - c) Symbiosis**
  - d) Competition
- 286) ---- is any symbiotic relationship in which one symbionts lives within the tissues of the other.
- a) Endosymbiosis**
  - b) Ectosymbiosis
  - c) None
  - d) Both**
- 287) Bacterial endosymbionts that provide essential nutrients to about ---- of insects.
- a) 10%–13%
  - b) 12%–15%
  - c) 10%–17%
  - d) 10%–15%**
- 288) Symbiotic relationship in which the symbionts lives on the body surface of the host, including the inner surface of the digestive tract or the ducts of exocrine glands.
- a) Endosymbiosis
  - b) Exosymbiosis**
  - c) None
  - d) Both
- 289) Barnacles are
- a) Ectosymbionts**
  - b) Endosymbionts
- 290) -----is a relationship between two organisms where one receives a benefit or benefits from the other and the other is not affected by it
- a) Mutualism
  - b) Commensalism**
  - c) Symbiosis
  - d) Competition

- 291) Remora and pilot fish feed on the leftovers of their hosts' meals. They show
- a) Mutualism
  - b) Commensalism**
  - c) Symbiosis
  - d) Competition
- 292) The species that benefits from the commensalism are called commensals.
- a) False
  - b) True**
- 293) Ancient DNA supports the hypothesis that ---- domestication preceded the emergence of agriculture
- a) Cat
  - b) Rabbits
  - c) Dog**
  - d) Mice
- 294) The dog was the first domesticant, and this animal was domesticated and widely established across -----.
- a) Indonesia
  - b) Eurasia**
  - c) Philippines
  - d) Japan
- 295) Epiphytic plants show
- a) Commensalism**
  - b) Mutualism
  - c) Symbiosis
- 296) ----- is the way two organisms of different species exist in a relationship in which each individual benefits from the activity of the other.
- a) Mutualism**
  - b) Commensalism
  - c) Symbiosis
  - d) Competition
- 297) Mutualism can be contrasted with?
- a) Intraspecific competition
  - b) Interspecific competition**
  - c) both
  - d) none
- 298) Relationship found between ungulates and bacteria within their intestines is.
- a) Mutualism**
  - b) Commensalism
  - c) Symbiosis
  - d) Competition
- 299) More than --- of land plants rely on mycorrhizal relationships with fungi.
- a) 49%
  - b) 48%**

- c) 50%
  - d) 60%
- 300) Oxpeckers or tickbirds feed on ticks that infest large, thick skinned animals it shows
- a) Parasitism
  - b) Mutualism**
  - c) Commensalism
- 301) Clownfish gains protection and food by living among deadly stinging sea anemones, it is an example of
- a) Parasitism
  - b) Mutualism**
  - c) Commensalism
- 302) A mutualistic association between algae and fungus in which both the partners get benefits from each other is known as ----.
- a) Lichen**
  - b) Ectomycorrhizas,
  - c) Mycorrhizas
  - d) Endomycorrhizas
- 303) How many growth form of lichens are?
- a) 4
  - b) 5
  - c) 3**
  - d) 2
- 304) ----- is a habit of some types of algae and lichens in which the plant grows tightly appressed to a substrate forming a biological layer of the adhering organism.
- a) Crustose**
  - b) Foliose
  - c) Fruticose
- 305) --- is found on rocks and tree bark.
- a) Crustose**
  - b) Foliose
  - c) Fruticose
- 306) A ----is one of a variety or lichens, which are complex organisms that arise from the symbiotic relationship between fungi and a photosynthetic partner, typically algae.
- a) Foliose lichen**
  - b) Crustose
  - c) Fruticose
- 307) The lifespan of foliose averages between?
- a) 30 and 70 years
  - b) 20 and 80 years
  - c) 40 and 60 years
  - d) 30 and 60 years**
- 308) Lichens develop quite slowly with recorded growth rates of ---depending on the species.
- a) 3.01-27mm/year

- b) **0.01-27mm/year**  
c) 0.01-29mm/year
- 309) ---is a form of lichen fungi that is characterized by a coral-like shrubby or bushy growth structure.  
a) Foliose lichen  
b) Crustose  
c) **Fruticose**
- 310) Which lichen is composed of a thallus and a holdfast?  
a) Foliose lichen  
b) Crustose  
c) **Fruticose**
- 311) What are the benefits of Lichens?  
a) Bio indicators and ecosystem sustainability  
b) Medicines and animal food  
c) Dye for clothes  
d) **All of above**
- 312) A ---is a symbiotic association composed of a fungus and roots of a vascular plant  
a) **Mycorrhizae association**  
b) Symbiotic relationship  
c) Lichen
- 313) How many types of Mycorrhizae are present?  
a) 2  
b) **3**  
c) 4  
d) 5
- 314) Mycorrhizas whose hyphae enter into the plant cells, producing structures that are either balloon-like (vesicles) or dichotomously branching invaginations (arbuscules) are called  
a) Ectomycorrhizas  
b) **Arbuscular mycorrhizas**  
c) both  
d) none
- 315) ---are typically formed between the roots of around 10% of plant families.  
a) **Ectomycorrhizas**  
b) Arbuscular mycorrhizas  
c) both  
d) none
- 316) A simple intraradical (grow in cells) phase, consisting of dense coils of hyphae in the outermost layer of root cells are  
a) Ectomycorrhizas  
b) Arbuscular mycorrhizas  
c) both  
d) **Ericoid mycorrhizas**

- 317) -----is a mutualistic association between nitrogen fixing bacteria and roots of plants of legume family.
- a) **Root nodule**
  - b) Ectomycorrhizas
  - c) Arbuscular mycorrhizas
  - d) All
- 318) Under nitrogen-limiting conditions, capable plants form a symbiotic relationship with a host-specific strain of bacteria known as-----.
- a) Rhizoids
  - b) **Rhizobia**
  - c) Ectoparasite
  - d) None
- 319) Free living aerobic bacteria is ----
- a) **Azobactor**
  - b) Clostridium
  - c) Thiobacillus
  - d) Nostoc
- 320) Free living anaerobic bacteria
- a) Azobactor
  - b) **Clostridium**
  - c) Thiobacillus
  - d) Nostoc
- 321) Photosynthetic bacteria
- a) **Cholorobium**
  - b) Clostridium
  - c) Thiobacillus
  - d) Nostoc
- 322) Chemosynthetic bacteria
- a) Cholorobium
  - b) Clostridium
  - c) **Thiobacillus**
  - d) Nostoc
- 323) Blue green algae
- a) Cholorobium
  - b) Clostridium
  - c) Thiobacillus
  - d) **Nostoc**
- 324) Non-mutual symbiotic relationship between species, where one species, the parasite, benefits at the expense of the other, the host.
- a) Camouflage
  - b) Mutualism
  - c) **Parasitism**
  - d) All

- 325) Parasites that live on the outside of the host, either on the skin or the outgrowths of the skin, are called -----
- a) **Ectoparasites**
  - b) Endoparasites
  - c) Both
- 326) Intracellular parasites, such as protozoa, bacteria or viruses, tend to rely on a third organism, which is generally known as ---
- a) **Vector**
  - b) Controller
  - c) Parasite
  - d) None
- 327) The definitive or final host is the host that harbors the -----of the parasite.
- a) Dormant stage
  - b) Inactive stage
  - c) **Sexual stages**
  - d) Recombinant stage
- 328) It occurs when an animal's color patterns help hide the animal, or a developmental stage, from another animal.
- a) **Camouflage**
  - b) Aposematic coloration
  - c) Mimicry
  - d) All
- 329) -----is a kind of camouflage common in frog and toad eggs.
- a) Camouflage
  - b) Aposematic coloration
  - c) Mimicry
  - d) **Counter shading**
- 330) When an animal takes on color patterns in its environment to prevent the animal from being seen by other animals.
- a) **Cryptic coloration**
  - b) Camouflage
  - c) Aposematic coloration
  - d) Mimicry
- 331) Some animals that protect themselves by being dangerous or distasteful to predators advertise their condition by conspicuous coloration.
- a) Cryptic coloration
  - b) Camouflage
  - c) **Aposematic coloration**
  - d) Mimicry
- 332) When a species resembles one, or sometimes more than one, other species and gains protection by the resemblance
- a) Cryptic coloration
  - b) Camouflage

- c) Aposematic coloration
  - d) **Mimicry**
- 333) Penguins are another example that exhibit phenomenon of
- a) Cryptic coloration
  - b) Camouflage
  - c) Aposematic coloration
  - d) **Counter shading**
- 334) Biologist **Edward O. Wilson** described two rules based on coloration.
- a) If it is small and strikingly beautiful, it is probably poisonous.
  - b) If it is strikingly beautiful and easy to catch, it is probably deadly.
  - c) If it is beautiful but not much attractive it is deadly
  - d) a and c
  - e) b and c
  - f) a and b
- 335) ----is a similarity of one species to another that protects one or both.
- a) Cryptic coloration
  - b) Camouflage
  - c) Aposematic coloration
  - d) **Mimicry**
- 336) ----involves using the toxic properties of chemical substances as weapons.
- a) **Chemical warfare**
  - b) Chemical reaction
  - c) Chemical explosions
  - d) All of above
- 337) When attacked, some species of squid and octopus emit clouds of black ink, allowing them to escape by confusing their predators. It is an example of?
- a) **Chemical warfare**
  - b) Chemical reaction
  - c) Chemical explosions
  - d) All of above
- 338) ----are very large ecological areas on the earth's surface, with fauna and flora adapting to their environment.
- a) Ecosystem
  - b) **Biomes**
  - c) Community
  - d) Ecosystem
- 339) Biomes are often defined by abiotic factors such as
- a) Climate
  - b) Relief
  - c) Geology
  - d) Soils
  - e) Vegetation
  - f) **All of above**

- 340) There are --- major categories of biomes on earth.  
a) 2  
b) 3  
c) 4  
d) 5
- 341) Oceans, coral reefs and estuaries, they come in categories of marine biomes or freshwater biomes?  
a) Freshwater  
b) **Marina**  
c) both  
d) none
- 342) There are --- main biomes that make up Forest Biomes.  
a) 1  
b) 2  
c) **3**  
d) 4
- 343) There are--- main types of grassland biomes.  
a) 1  
b) **2**  
c) 3  
d) 4
- 344) There are --- major tundra biome.  
a) 1  
b) **2**  
c) 3  
d) 4
- 345) Desert biomes make ---of earth's biome.  
a) **19%**  
b) 23%  
c) 12%  
d) 33%
- 346) The process that involves the cyclic movement of nutrients between biotic and abiotic environment of earth is termed as ---  
a) **Biogeochemical cycle**  
b) Biological system  
c) Biological weather  
d) Biology
- 347) Approximately ---of living matter is made of oxygen, carbon, nitrogen, and hydrogen.  
a) **97%**  
b) 99%  
c) 44%  
d) 33%

- 348) Sedimentary cycle's elements are --- than those of gases.  
a) **Less abundant**  
b) More abundant  
c) Fewer  
d) Greater
- 349) CO<sub>2</sub> gas makes up --- of the volume of the atmosphere.  
a) 0.0338%  
b) 0.0038%  
c) **0.038%**  
d) 0.0138%
- 350) ---is a key component of nature's thermostat.  
a) **CO<sub>2</sub>**  
b) NO<sub>2</sub>  
c) H<sub>2</sub>O  
d) SO<sub>2</sub>
- 351) Aerobic respiration occurs in  
a) Producers  
b) Consumers  
c) Decomposers  
d) **All**
- 352) Specialized soil bacteria convert most of the Ammonia (NH<sub>3</sub>) and Ammonium ions (NH<sub>4</sub><sup>+</sup>) in soil to nitrate ions (NO<sub>3</sub><sup>-</sup>). This process is called  
a) **Nitrification**  
b) Nitrification  
c) Ammonification  
d) Nitrogen fixation
- 353) Circulates through water, the earth's crust, and living organisms.  
a) Nitrogen  
b) Carbon dioxide  
c) Air  
d) **Phosphorous**
- 354) Does not includes atmosphere  
a) **Phosphorous cycle**  
b) Nitrogen cycle  
c) Water cycle  
d) Sulfur cycle
- 355) Phosphorous is a component of biologically important molecules such as  
a) Nucleic acids  
b) Energy transfer molecules (ADP and ATP)  
c) Vertebrate bones  
d) Teeth  
e) **All of above**
- 356) ---- is often the limiting factor for plant growth

- a) Sulfur
  - b) Carbon
  - c) Water
  - d) Phosphate**
- 357) Sulfur circulates through the ---- in the sulfur cycle.
- a) Biosphere**
  - b) Atmosphere
  - c) Ecosystem
  - d) Biomes
- 358) Marine algae produce large amounts of.
- a) Dimethyl sulfide**
  - b) Carbon dioxide
  - c) Oxygen
  - d) All of above
- 359) Changes liquid water into water vapor in the atmosphere.
- a) Evaporation**
  - b) Transpiration
  - c) Precipitation
  - d) All
- 360) Incoming solar energy causes ---- of water from the oceans, lakes, rivers, and soil.
- a) Evaporation**
  - b) Transpiration
  - c) Precipitation
  - d) All
- 361) About --- of water vapor in the atmosphere comes from the oceans, which cover almost three-fourths of the earth's surface.
- a) 84%**
  - b) 87%
  - c) 67%
  - d) 77%
- 362) Over land, about --- of the water that reaches the atmosphere evaporates from the surfaces of plants through a process called transpiration.
- a) 90%**
  - b) 77%
  - c) 99%
  - d) 89%
- 363) Returns the water back to earth's surface.
- a) Evaporation
  - b) Transpiration
  - c) Precipitation**
  - d) All
- 364) Point sources are single, identifiable sources. Which of the following is its example?

- a) Smokestack of a coal-burning power or industrial plant
  - b) The drainpipe of a factory.
  - c) The exhaust pipe of an auto mobile
  - d) **All**
- 365) Nonpoint sources are dispersed and often difficult to identify. Which of the following is its example?
- a) Pesticides blown from the land into the air
  - b) The runoff of fertilizers and pesticides from farmlands
  - c) Lawns, gardens, and golf courses into streams and lakes.
  - d) **All of above.**
- 366) It is much easier and cheaper to identify and control or prevent pollution from—?
- a) **Point sources**
  - b) Non point sources
  - c) Both
  - d) None
- 367) human sewage and newspapers are
- a) **Biodegradable pollutants**
  - b) Non-biodegradable pollutants
  - c) Both
- 368) Lead, mercury, and arsenic are
- a) **Non-biodegradable pollutants.**
  - b) Biodegradable pollutants
  - c) Both
- 369) Organic pollutants or bio stimulants, primarily from agricultural fertilizers and sewage waste, have a major impact on ----environments.
- a) **Aquatic**
  - b) Terrestrial
  - c) Both
- 370) When the excessive nutrients enter aquatic systems, they stimulate.
- a) **Plant growth.**
  - b) Algal growth.
  - c) Fungal growth.
- 371) Rapid phytoplankton growth or algal blooms creates ---- that is, low-oxygen, less than 2 mg/liter or even -----environments in which few organisms can survive.
- a) Anoxic
  - b) **Hypoxic, anoxic**
  - c) Anoxic, hypoxic
- 372) Large concentrations of algae also reduce water-----, preventing light from reaching the bottom and reducing the growth of ----.
- a) Quality, sea grasses
  - b) Quantity, sea grasses
  - c) **Clarity, sea grasses**

- 373) A long-term increase in excess nutrients into an ecosystem causes higher biological activity this is known as
- a) Harmful algal bloom
  - b) Eutrophication
  - c) **Both a and b**
- 374) Eutrophication -----the oxygen content rapidly.
- a) **Decreases**
  - b) Increases
  - c) Lowers
  - d) Enhances
- 375) What are the sources of water pollution?
- a) Oil spills
  - b) Agricultural runoff
  - c) Suspended or particulate matter
  - d) **All of above**
- 376) What are the effects of water pollution?
- a) Diseases
  - b) Death of living organisms
  - c) Loss of vegetation
  - d) Habitat loss
  - e) **All of above**
- 377) Air pollutants can be in the form of
- a) Solid particles
  - b) Gases
  - c) Liquid droplets
  - d) **All given forms**
- 378) Burning fossil fuels releases sulfur dioxide and nitrogen oxides into the atmosphere. Sulfur dioxide and water combine to produce sulfuric acid, which falls as.
- a) Acid rain
  - b) Acid deposition
  - c) Pollutants
  - d) **Both a and b**
- 379) Acid deposition---the pH of lakes.
- a) Increases
  - b) **Lowers**
- 380) Carbon dioxide released from burning fuels accumulates in the atmosphere and prevent the escape of heat coming from earth.
- a) **Greenhouse effect.**
  - b) Air pollution
  - c) Global warming
- 381) Carbon dioxide reflects solar radiation back to the earth. This reflection of solar radiation is predicted to cause an increase in world temperature, polar ice caps to melt, and ocean levels to rise.

- a) Greenhouse effect
  - b) **Global warming**
  - c) Air pollution
- 382) Ozone layer depletion causes the chance of.
- a) Eye cancer
  - b) **Skin cancer**
  - c) Mouth cancer
- 383) The accumulation of matter in food webs is called
- a) **Biological magnification**
  - b) Biological identification
  - c) Biological classification
- 384) Cars, trains, airplanes, shipping and industry (construction or factory) are the leading sources of.
- a) Air pollution
  - b) Water pollution
  - c) Soil pollution
  - d) **Noise pollution**
- 385) Animals rely on hearing to
- a) Communicate
  - b) Avoid predators
  - c) Obtain food
  - d) **All of above**
- 386) Effect of noise pollution
- a) Hearing loss
  - b) Loss of communication
  - c) Physiological effects because of increased heart rate and metabolism
  - d) **All given above**
- 387) Noise from sonar or explosives causes marine mammals to.
- a) **Dive deeper**
  - b) Come on the surface
  - c) Die easily
- 388) On long, deep dives, more --- enters the blood from the lungs in the form of bubbles
- a) Carbon
  - b) **Nitrogen**
  - c) oxygen
  - d) hydrogen
- 389) On long, deep dives, more nitrogen enters the blood from the lungs in the form of bubbles and kills the animal. In humans, this illness is known as the.
- a) **Bends**
  - b) cataract
  - c) asthma
  - d) lungs cancer
- 390) Some short term global changes occur during---.

- a) **Volcanic eruption**
  - b) Floods
  - c) Droughts
  - d) Succession
- 391) In ---Mount Pinatubo in the Philippines erupted, producing huge quantities of gas and ash reaching the stratosphere resulting in a worldwide -----.
- a) 1992, warming
  - b) **1991, cooling**
  - c) 1990, warming
  - d) 1999, cooling
- 392) It is estimated that ----of SO<sub>2</sub> gas erupted into the stratosphere.
- a) 20,000,00 tons
  - b) 20,0,000 tons
  - c) 20,000 tons
  - d) **20,000,000 tons**
- 393) H<sub>2</sub>SO<sub>4</sub> reflected ----of the incoming solar radiation back into the space, reducing the mean global temperature at the surface about---.
- a) 3 to 4 percent, 0.8°C
  - b) **2 to 4 percent, 0.5°C**
  - c) 2 to 3 percent, 0.4°C
- 394) Greenhouse gases allow mostly visible light and some IR radiation and UV radiation from the sun to pass through the ----.
- a) **Atmosphere**
  - b) Stratosphere
  - c) Troposphere
  - d) Ozone layer
- 395) Growing crops release --- into the atmosphere.
- a) Carbon dioxide
  - b) Methane
  - c) Nitrous oxide
  - d) **All**
- 396) Another type of short-term climate change is the so-called El Nino effect, which occurs every --- when the typical weather pattern in the Pacific Ocean breaks down.
- a) 2 to 8 years
  - b) 3 to 7 years
  - c) **2 to 7 years**
- 397) This change causes warm surface water to move toward South and Central America, reversing the directions of some currents and placing warm water along the coasts.
- a) Global warming
  - b) Greenhouse effect
  - c) **El Nino effect**
- 398) The warm water El Nino brings to the coasts of North and South America reduces the upwelling of cold, deep water, affecting the cold water---.

- a) **Fisheries**
  - b) Snakes
  - c) Birds
  - d) Mammals
- 399) Causes drastic and irreversible changes both in the upper atmosphere and within the planet thus affecting it's every component including land, water, air.
- a) Greenhouse effect
  - b) **Global warming**
  - c) Both
- 400) Solar ultraviolet radiation breaks apart one oxygen molecule to produce two oxygen atoms each of these highly reactive atoms combines with an oxygen molecule to produce an ozone molecule. This process occurs in ----.
- a) **Stratosphere ozone**
  - b) Tropospheric ozone
  - c) Both
  - d) None
- 401) Nitrogen compounds released from automobiles is broken down by sunlight into nitrogen oxide. It is then converted in to nitrogen monoxide releasing an oxygen molecule that combines with oxygen in the atmosphere to form Ozone. This is
- a) Stratosphere ozone
  - b) **Tropospheric ozone**
  - c) Both
  - d) None
- 402) Ozone abundances in the stratosphere and troposphere are determined by
- a) Amount of reactive gases
  - b) Sunlight intensity
  - c) Temperature
  - d) **All**
- 403) There should be a balance between production and degradation of ozone in the.
- a) **Stratosphere**
  - b) Troposphere
  - c) Both
  - d) None
- 404) Effects of Ozone depletion
- a) Skin cancer
  - b) Snow-blindness
  - c) Cataract
  - d) Low crop production
  - e) **All given**
  - f) None
- 405) Irreparable computers and other electronic goods are known as
- a) Solid wastes
  - b) **E-wastes**

- c) None
  - d) Both
- 406) Which of the following is more important to protect?
- a) preventing premature extinction of species
  - b) protect threatened habitats
  - c) protect ecosystem
  - d) b and c**
- 407) Humans already capture more than a ---of the earth's terrestrial productivity and more than -----of the world's fresh water.
- a) **Third, half**
  - b) Forth, second half
  - c) Half, half
  - d) None
- 408) As food production has doubled in the last ----years.
- a) 36
  - b) 35**
  - c) 30
  - d) 10
- 409) Now we release more ----- into the environment than all natural sources combined.
- a) Nitrogen and SO<sub>2</sub>
  - b) Sulfur and phosphorus
  - c) Nitrogen and phosphorus**
  - d) nitrogen and CO<sub>2</sub>
- 410) Which of the following actions can helpful in attaining a stable population?
- a) Improve the educational and political status of women.
  - b) Improve the survival and health of children
  - c) Provide easy access to family planning resources
  - d) All of above**
- 411) The term biodiversity was used by ---to describe the combined diversity at all the levels of biological organization.
- a) Edward Wilson**
  - b) Ernst Heckle
  - c) Wittaker
- 412) Biodiversity is a vital----- resource.
- a) Nonrenewable.
  - b) Renewable**
  - c) None
- 413) The variety of life on Earth at all its levels, from genes to ecosystems, and the ecological and evolutionary processes that sustain it. It is called ----.
- a) Biodiversity**
  - b) Biology
  - c) Bio magnification
  - d) Biological classification

- 414) Scientists have identified about ----species  
a) 2.8 million  
b) 1.7 million  
**c) 1.8 million**
- 415) The identified species include almost -----species of insects.  
a) **1 million**  
b) 2 million  
c) 1.5 million  
d) 0.5 million
- 416) -----plant species are identified.  
a) 270,0000  
b) 2700  
**c) 270,000**  
d) 27000
- 417) Vertebrate animal species identified  
**a) 45,000**  
b) 45,0000  
c) 45,00000
- 418) Biodiversity vary with respect to changes in the  
a) Geology  
b) Natural selection  
c) Species evolution  
d) Sea level  
**e) All of above**
- 419) Refers to number and abundance of species present in different communities.  
a) Gene diversity  
**b) Species diversity**  
c) Population diversity  
d) Community diversity
- 420) Species diversity is the variation in the number and phylogenetic diversity or evolutionary relatedness of species present in an area.  
a) Gene diversity  
**b) Species diversity**  
c) Population diversity  
d) Community diversity
- 421) More than ----of all the species recorded are animals.  
a) 80%  
**b) 70%**  
c) 30%  
d) 99%
- 422) Plants including algae, fungi, bryophytes, gymnosperms and angiosperms comprise no more than ----of the total.  
a) 80%

- b) **22%**
  - c) 30%
  - d) 99%
- 423) Among animals, insects are the most species-rich taxonomic group, making up more than ---of the total.
- a) 80%
  - b) **70%**
  - c) 30%
  - d) 99%
- 424) The number of fungi species in the world is more than the combined total of the species of
- a) Fishes, amphibians, reptiles
  - b) Fishes, reptiles
  - c) **Fishes, amphibians, reptiles, mammals**
  - d) Fishes, reptiles and mammals
- 425) Refers to variation in the interdependence of biotic communities and the abiotic aspects of the environments in which the biotic communities are found.
- a) Gene diversity
  - b) Species diversity
  - c) Population diversity
  - d) **Ecosystem diversity**
- 426) Ecosystem is a storehouse of
- a) Genetic diversity
  - b) Species diversity
  - c) Population diversity
  - d) **Both a and b**
- 427) Ecosystems develop in response to local conditions, which are influenced by
- a) Climate patterns
  - b) Soil types
  - c) Topography
  - d) **All given**
- 428) Variety of genetic material within a species or a population.
- a) **Genetic diversity**
  - b) Species diversity
  - c) Population diversity
  - d) Both a and b
- 429) Vital to the sustainability of life on earth.
- a) **Genetic diversity**
  - b) Species diversity
  - c) Population diversity
  - d) Both a and b
- 430) Genetic diversity will help the individuals to survive under
- a) Climate change
  - b) Drought

- c) Fire
  - d) Famine
  - e) **All given**
- 431) Who created a system to describe biodiversity over different spatial scales.
- a) Edward Wilson
  - b) Ernst Heckle
  - c) **Robert Whittaker (1972)**
  - d) Albert Einstein
- 432) Diversity within a particular area or ecosystem, and it is usually expressed by the number of species in that ecosystem.
- a) **Alpha diversity**
  - b) Beta diversity
  - c) Gemma diversity
- 433) Alpha diversity is equivalent to measuring the species ----of an area.
- a) Abundance
  - b) Evenness
  - c) **Richness**
- 434) If we want to monitor aquatic biodiversity on the offshore side of a fringing coral reef in the Ocean. Then the ----of this reef is measured by the total number of species found in the area that we are monitoring.
- a) **Alpha diversity**
  - b) Beta diversity
  - c) Gemma diversity
- 435) We can compare the degree of species change as we move along the transect from one ecosystem to the next in
- a) Alpha diversity
  - b) **Beta diversity**
  - c) Gemma diversity
- 436) Comparison of diversity between ecosystems, usually measured as the amount of species change between the ecosystems.
- a) Alpha diversity
  - b) **Beta diversity**
  - c) Gemma diversity
- 437) ---is a measure of the overall diversity within a large region, and so in some respects it is an expanded version of alpha diversity.
- a) Alpha diversity
  - b) Beta diversity
  - c) **Gemma diversity**
  - d) None
- 438) Undisturbed forests maintain the rainfall in their immediate environment by contributing to the---.
- a) **Hydrological cycle**
  - b) Nitrogen cycle

- c) Sulfur cycle
- d) Maintenance of ecosystem
- 439) Habitat loss is the permanent ---of land to other uses.
  - a) Retention
  - b) **Conservation**
  - c) Destruction
  - d) Dispersal
- 440) Today, ---activities are the major cause of habitat loss and fragmentation throughout the world.
  - a) Human
  - b) **Agricultural**
  - c) Both
- 441) Mangroves, sea grasses, salt marshes, coral reefs, kelp forests, and rocky shorelines are fragmented by natural forces such as
  - a) Wave action
  - b) Currents and Tides
  - c) Storm surge and Human activities
  - d) **All of above**
- 442) Species whose populations have expanded dramatically and out-compete or displace native species, potentially threatening the structure and function of intact ecosystems.
  - a) Dormant species
  - b) Indigenous species
  - c) **Invasive species**
  - d) None of above
- 443) The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than ---species of cichlid fish in the lake.
  - a) 300
  - b) **200**
  - c) 400
  - d) 100
- 444) The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous ---in our rivers
  - a) **Catfish**
  - b) Dog
  - c) Birds
  - d) Lizards
- 445) Brown tree snake was introduced inadvertently on many Pacific islands and subsequently caused the extinction of a number of
  - a) Native birds
  - b) Bats
  - c) Lizards
  - d) **All of them**
- 446) There appear to be three major stages in the process of invasion by exotic species.

- a) **Dispersal, Establishment, Integration**
  - b) Establishment, dispersal, integration
  - c) Integration, dispersal, establishment
- 447) Overexploitation of resources is a critical problem in.
- a) **Conservation**
  - b) Biodiversity
  - c) Ecosystem
- 448) Closely linked and plays an increasing role in the loss of biodiversity.
- a) Conservation
  - b) Biodiversity
  - c) Ecosystem
  - d) **Over exploitation**
- 449) Ranges from commercial activities such as logging operations or trade in endangered species to subsistence hunting.
- a) Indirect exploitation
  - b) **Direct exploitation**
  - c) None
  - d) Both
- 450) Encompasses the unintentional mortality of non target species such as fish or turtles killed as by-catch in fishery operations.
- a) **Indirect exploitation**
  - b) Direct exploitation
  - c) None
  - d) Both
- 451) Chlorinated hydrocarbons, such as are renowned for their toxic effects on the environment.
- a) DDT (dichloro-diphenyl-trichloro ethane)
  - b) PCB (polychlorobenzene)
  - c) **Both a and b**
- 452) A phenomenon called bio magnification is well known for.
- a) mercury
  - b) DDT
  - c) **mercury and DDT**
  - d) none
- 453) Refers to increase in concentration of the toxicant at successive trophic levels.
- a) **Bio magnification**
  - b) Bio degradation
  - c) Bio indication
- 454) According to an estimate, almost ---forests have been lost in the tropics, compared to only ----in the temperate region.
- a) 49%, 3%
  - b) 50%, 5%
  - c) **40%, 1%**
  - d) 80%, 3%

- 455) Humans are either directly or indirectly exploiting about --of the earth's net primary production.
- a) 49%
  - b) 50%
  - c) **40%**
- 456) Tropical rain forests cover only --- of the earth's land surface, but they contain more than -----of the world's species.
- a) 6%, 50%
  - b) 5%, 60%
  - c) **7%, 50%**
  - d) 9%, 49%
- 457) About ---- an area larger than Costa Rica is being cleared each year of tropical rain forest.
- a) 76,00 km<sup>2</sup>
  - b) 76,0000 km<sup>2</sup>
  - c) **76,000 km<sup>2</sup>**
  - d) 76,00000 km<sup>2</sup>
- 458) R.H. Whittaker ---proposed a Five Kingdom classification.
- a) **1969**
  - b) 1869
  - c) 1979
  - d) 1968
- 459) The main criteria used for classification is
- a) Thallus organization
  - b) Mode of nutrition
  - c) Reproduction
  - d) Phylogenetic relationships
  - e) **All of above**
- 460) Bacteria are the sole members of the Kingdom
- a) Protista
  - b) **Monera**
  - c) Fungi
  - d) Animalia
- 461) Vibrium: These are ---shaped bacteria.
- a) **Comma**
  - b) Cylinder
  - c) Spherical
- 462) All members of the Archeobacteria inhabit ---- environments.
- a) **Anaerobic**
  - b) Aerobic
  - c) Both
- 463) These types of bacteria are found in extreme salty areas like great salt lake and dead sea.
- a) Psychrophiles
  - b) **Halophiles**

- c) Vibrium  
d) Thermophiles
- 464) *Salinibacter ruber* is an example of which kind of bacteria  
a) Psychrophiles  
**b) Halophiles**  
c) Vibrium  
d) Thermophiles
- 465) *Thermus aquaticus* and *Thermococcus litoralis* are example of.  
a) Psychrophiles  
b) Halophiles  
c) Vibrium  
**d) Thermophiles**
- 466) *Listeria monocytogenes* is a ---  
**a) Psychrophiles**  
b) Halophiles  
c) Vibrium  
d) Thermophiles
- 467) ---are also named as cold loving bacteria.  
**a) Psychrophiles**  
b) Halophiles  
c) Vibrium  
d) Thermophiles
- 468) These are found in swap and marshy areas  
**a) Methanogens**  
b) Chemosynthetic bacteria  
c) Heterobacteria  
d) Methanobacterium
- 469) ---present in the gut of ruminants are responsible for the production of biogas from the dung of these animals.  
a) Chemosynthetic bacteria  
b) Heterobacteria  
**c) Methanobacterium**
- 470) Archeobacteria differ from other bacteria in having a different ----structure  
a) Cell membrane  
b) Cytoplasm  
c) Organelles  
**d) Cell wall**
- 471) *Nostoc* and *Anabaena* are examples of.  
a) **Cyanobacteria**  
b) Chemosynthetic bacteria  
c) Heterobacteria
- 472) Some of the cyanobacteria can fix atmospheric nitrogen in specialized cells called.  
**a) Heterocysts**

- b) Cysts
  - c) Cells
  - d) None
- 473) ----bacteria oxidize various inorganic substances such as nitrates, nitrites and ammonia and use the released energy for their ATP production.
- a) **Chemosynthetic**
  - b) Cyanobacteria
  - c) Photosynthetic
  - d) Both
- 474) Which of the following is characteristics of heterobacteria?
- a) Synthesize curd from milk
  - b) Antibiotic production
  - c) Nitrogen fixation
  - d) Pathogens
  - e) **All of above**
- 475) Bacteria reproduce through
- a) Fission
  - b) Spore formation
  - c) **Both**
- 476) What are the characteristics of mycoplasma?
- a) Lack cell wall
  - b) Smallest living cells
  - c) Survive without oxygen
  - d) Pathogenic
  - e) **All of above**
- 477) Protista was first used by Ernst Haeckel in the year.
- a) 1888
  - b) 1986
  - c) **1886**
- 478) What are the characteristics of protists
- a) Unicellular eukaryotes
  - b) Flagella or cilia
  - c) Cyst formation
  - d) **Given all**
- 479) Protists are classified in --- main classes.
- a) 2
  - b) 3
  - c) 4
  - d) 5
- 480) Protists reproduce asexually by the process of ---and sexually by a process involving---.
- a) **Cell fusion, zygote formation**
  - b) Zygote formation, cell fusion
  - c) Cell fission, zygot formation

- d) Zygot fusion, cell fission
- 481) Amoeba, Paramecium, Euglena, Diatoms, Dinoflagellates these are examples of.
- a) **Protists**
  - b) Animals
  - c) Plants
  - d) Fungi
- 482) ---are plant like protists. They are mostly unicellular or colonial organisms which are found in fresh and salt water throughout the world.
- a) **Chrysophytes**
  - b) Dinoflagellates
  - c) Euglenoids
  - d) Slime molds
- 483) Examples of Chrysophytes
- a) Diatoms
  - b) Golden brown algae
  - c) Yellow-green algae
  - d) **All given**
- 484) What are the Characteristics of Chrysophytes?
- a) Unicellular eukaryotes
  - b) Freshwater and saltwater habitat
  - c) Planktons
  - d) Photosynthesis
  - e) **All of above**
- 485) The shells of fossil diatoms form thick deposits on the sea floor known as
- a) **Diatomaceous earth**
  - b) Diatoms
  - c) Dinoflagellate
- 486) Characters of diatoms are shells with silica
- a) Photosynthesis
  - b) Major producers in oceans
  - c) Water filters
  - d) Diatomaceous earth
  - e) **All**
- 487) Diatomaceous earth is used in
- a) Polishing
  - b) Filtration of oils
  - c) Syrups
  - d) **All of above**
- 488) Dinoflagellates store food in the form of.
- a) **Starch like compounds**
  - b) Proteins like compounds
  - c) Carbohydrates like compounds
- 489) Dinoflagellate cell wall has stiff cellulose plates on the----.

- a) Inner surface
  - b) Outer surface**
  - c) Middle surface
- 490) Under favorable conditions Dinoflagellates populations may reach ---organisms per liter of water
- a) 650 million
  - b) 90 million
  - c) 60 million**
- 491) Euglenoids make up a large group of common microorganisms whose known species range is between ---- species.
- a) 740-990
  - b) 750-800
  - c) 760-900
  - d) 750-900**
- 492) Euglenoids have a protein rich layer called ---which makes their body flexible.
- a) Ventricle
  - b) Pellicle**
  - c) Cell wall
  - d) Protoplasm
- 493) Euglenoids have
- a) Animals like structure
  - b) Plant like structure
  - c) Both**
- 494) ----are a kind of protists that aggregates into colonies and ingest fungi, bacteria, protozoa, other micro-organisms, and dead and decaying organic matter.
- a) Slime moulds**
  - b) Euglenoids
  - c) Protozoans
  - d) Protists
- 495) Under suitable conditions, they form an aggregation called ----which may grow and spread over several feet.
- a) Mycelium
  - b) Plasmodium**
  - c) Sporangium
  - d) None
- 496) All ----are heterotrophs and live as predators or parasites
- a) Protozoans**
  - b) Euglenoids
  - c) Protists
  - d) Fungi
- 497) These organisms live in fresh water, sea water or moist soil. They move and capture their prey by putting out pseudopodia are
- a) Amoeboid protozoans**

- b) Flagellated protozoans
  - c) Ciliated protozoans
  - d) Sporozoans
- 498) Which of the following cause sleeping sickness
- a) Amoeboid protozoans
  - b) Flagellated protozoans**
  - c) Ciliated protozoans
  - d) Sporozoans
- 499) Paramecium is an example of
- a) Amoeboid protozoans
  - b) Flagellated protozoans
  - c) Ciliated protozoans**
  - d) Sporozoans
- 500) Plasmodium is an example of
- a) Amoeboid protozoans
  - b) Flagellated protozoans
  - c) Ciliated protozoans
  - d) Sporozoans**
- 501) Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called
- a) Coenocytic hyphae**
  - b) Mycelium
  - c) Sporangium
  - d) None
- 502) The cell walls of fungi are composed of.
- a) Chitin
  - b) Polysaccharides
  - c) Both**
- 503) Mucor, Rhizopus (the bread mould), Albugo (the parasitic fungi on mustard) these are examples of
- a) Phycomycetes**
  - b) Actinomycetes
  - c) Basidiomycetes
  - d) Deuteromycetes
- 504) Reproduce asexually, aseptate, coenocytic, endogenous spore formation are characters of
- a) Phycomycetes**
  - b) Actinomycetes
  - c) Basidiomycetes
  - d) Deuteromycetes
- 505) Penicillium and Saccharomyces are example of
- a) Phycomycetes
  - b) Actinomycetes**
  - c) Basidiomycetes

- d) Deutermycetes
- 506) Saprophytic, Decomposers, Parasitic or coprophilous (growing on dung), Mycelium is branched and septate are characters of
- a) Phycomycetes
  - b) Actomycetes**
  - c) Basidiomycetes
  - d) Deutermycetes
- 507) Aspergillus, Claviceps, Neurospora are examples of
- a) Phycomycetes
  - b) Actomycetes**
  - c) Basidiomycetes
  - d) Deutermycetes
- 508) Neurospora is used extensively in
- a) Biochemical work
  - b) Genetic work
  - c) Both**
- 509) Morels and truffles are ---- and are considered delicacies
- a) Nonedible
  - b) Edible**
  - c) Poisonous
  - d) Unhealthy
- 510) Agaricus (mushroom), Ustilago (smut), Puccinia (rust fungus are examples of
- a) Phycomycetes
  - b) Actomycetes
  - c) Basidiomycetes**
  - d) Deutermycetes
- 511) ----- help in mineral cycling
- a) Phycomycetes
  - b) Actomycetes
  - c) Basidiomycetes
  - d) Deutermycetes**
- 512) Alternaria, Colletotrichum , Trichoderma are examples of
- a) Phycomycetes
  - b) Actomycetes
  - c) Basidiomycetes
  - d) Deutermycetes**
- 513) Deuteromycetes reproduce through
- a) Asexual reproduction**
  - b) Sexual reproduction
  - c) Binary fission
  - d) None